

Name: G.Abhilash

Roll_No:2303A52240

PROMPT ENGINEERING TASK REPORT

Task 1: Cleaning Sensor Data

Prompt:

Give a function that filters out all the negative numbers from a list.**Explanation**

As the prompt becomes more specific, the calculator logic improves by handling operators correctly and preventing division by zero.

Code

```
def filter_negative_numbers(numbers):  
    """  
    Filters out all negative numbers from a list.  
  
    Args:  
    | numbers (list): List of numbers to filter  
  
    Returns:  
    | list: List containing only non-negative numbers  
    """  
    return [num for num in numbers if num >= 0]  
  
# Example usage:  
# numbers = [-5, 3, -1, 0, 8, -2, 10]  
# filtered = filter_negative_numbers(numbers)  
# print(filtered) # Output: [3, 0, 8, 10]
```

Output

```
Original array: [-5, 3, -1, 0, 8, -2, 10]  
Filtered array: [3, 0, 8, 10]
```

Task 2: String Character Analysis

Prompt

give a function that counts vowels, consonants, and digits in a string

Code

```
def count_string_components(text):  
    vowels = "aeiouAEIOU"  
    vowel_count = 0  
    consonant_count = 0  
    digit_count = 0  
  
    for char in text:  
        if char.isdigit():  
            digit_count += 1  
        elif char.isalpha():  
            if char in vowels:  
                vowel_count += 1  
            else:  
                consonant_count += 1  
  
    return vowel_count, consonant_count, digit_count
```

Output

```
Original array: [-5, 3, -1, 0, 8, -2, 10]  
Filtered array: [3, 0, 8, 10]  
String: 'Hello World 123!'  
Vowels: 3  
Consonants: 7  
Digits: 3
```

Task 3: Palindrome Check – Tool Comparison

Prompt

give a function that checks the palindrome

.

Code

```
def is_palindrome(text):  
    # Convert to lowercase and keep only alphanumeric characters  
    cleaned = ''.join(char.lower() for char in text if char.isalnum())  
  
    # Check if cleaned string equals its reverse  
    return cleaned == cleaned[::-1]
```

Output

```
'racecar' -> True  
'A man a plan a canal Panama' -> True  
'hello' -> False  
'Madam' -> True  
'12321' -> True  
'12345' -> False
```

Task 4: Code Explanation Using AI

Prompt

Explain the palindrome code

Code

```
Check if a given text is a palindrome.  
This function determines whether the input text reads the same forwards and backwards,  
ignoring case differences, spaces, and non-alphanumeric characters.  
Args:  
    text (str): The input string to check for palindrome property.  
Returns:  
    bool: True if the text is a palindrome, False otherwise.  
Example:  
    >>> is_palindrome("A man a plan a canal Panama")  
    True  
    >>> is_palindrome("race a car")  
    False  
    >>> is_palindrome("Was it a car or a cat I saw?")  
    True  
Note:  
    The function performs case-insensitive comparison and ignores all  
    non-alphanumeric characters (spaces, punctuation, etc.) when  
    determining if the text is a palindrome.
```